

***Prompt Engineering and LLM
Fundamentals***

PROMPT ENGINEERING PRACTICE BOOK

Introduction:

This is my practice book for Assignment #5. For eight use cases I run into regularly - things like writing emails, summarizing documents, or generating chatbot replies - I wrote three versions of the same prompt: a basic one, an improved one, and a final professional version. The goal was to actually see how much of a difference structure and specificity make to what an LLM gives back, instead of just assuming a "better" prompt helps

Prompts — By Use Case

Summarizing a Policy Document

Purpose: To create a short and accurate summary of a long policy document so stakeholders can quickly understand the most important information.

1. Basic Prompt

Give a summary of the given document.

Expected Output: A short overview (typically one or two paragraphs) covering the main purpose of the policy document.

2. Improved Prompt

Summarize the document by highlighting the important terms and key details.

Expected Output: This prompt gives concise summary, list of important policy points, highlighted important terms, more informative than the basic summary.

3. Professional Prompt

Summarize the policy document by extracting all key points. Organize the summary using headings and bullet points, include the page number for each major point, explain any technical or legal terms in simple language, and conclude with the document's overall purpose and key takeaways.

Expected Output: This prompt gives professional executive summary, all important policy points, page references for each major point, explanation of important terminology, better organized and easier to review

Comparison Note: Each version becomes better than the previous one. The Basic prompt gives only a simple summary. The Improved prompt also highlights important terms. The Professional prompt organizes the summary with headings, page numbers, and explains legal terms in simple language. This shows a clear improvement in prompt quality.

Extracting Key Points from a Report

Purpose: To identify and present the most important findings and recommendations from a report for quick review and decision-making.

1. Basic Prompt

Give important key points from this document.

Expected Output: A simple list of important points without any order or headings.

2. Improved Prompt

Extract important points in bullet point form and also mention the heading of each topic that is in the report.

Expected Output: The important points are grouped under their original headings, making the report easier to read.

3. Professional Prompt

Extract important points from this document, highlight the key details, organize these points with their headings, mention the page numbers containing important information, explain all important terms in detail, do not mention any other point that is not stated in the report and produce a structured and comprehensive summary.

Expected Output: A well-organized summary with headings, page numbers, highlighted key details, and explanations of important terms.

Comparison Note: The Basic prompt only lists important points. The Improved prompt organizes them under headings. The Professional prompt makes the summary more detailed by adding page numbers and explanations of important terms.

Generating Meeting Minutes

Purpose: To convert a meeting transcript into clear, well-organized meeting minutes that record discussions, decisions, and action items.

1. Basic Prompt

Write meeting minutes for the meeting that I attend today and give summary of the important details that I discussed today.

Expected Output: A simple paragraph describing what happened in the meeting.

2. Improved Prompt

Write meeting minutes. Write important details discussed in the meeting in bullet point form. Mark the technical terms that are in the points. Include the key decisions made during the meeting and list any action items.

Expected Output: Important points written in bullet form with technical terms included.

3. Professional Prompt

Now think of yourself as a professional assistant. Record the complete meeting minutes. Note all the important topics discussed during the meeting. Write all technical terms under appropriate headings. Include all important documents referenced during the meeting. Write in formal, professional English. Avoid personal opinions and interpretations. Provide a structured summary with proper headings and bullet points. Maintain consistent tense throughout.

Expected Output: A formal meeting minutes document with headings, technical terms, referenced documents, and consistent writing style.

Comparison Note: The Professional version is much better because it follows a proper meeting minutes format and keeps the writing formal and consistent.

Creating a Project Proposal

Purpose: To prepare a clear, well-structured project proposal that explains the project and helps stakeholders make informed approval decisions.

1. Basic Prompt

Write a project proposal on the topic of AI – THE NEW GEN ERA. Give all important details for this topic with their headings.

Expected Output: A simple proposal with basic headings.

2. Improved Prompt

Write a structured project proposal on the topic AI – THE NEW GEN ERA. Give its summary, objective, why it is needed, scope of work, challenges, pros and cons, recommendations, timelines, conclusions, and future scope. Write in structured form.

Expected Output: A well-structured proposal covering all required sections.

3. Professional Prompt

Write a project proposal as a business consultant. Give all the key details about the project. Include these sections in your proposal: Executive Summary, Problem Statement, Objectives, Scope of Work, Methodology, Tools and Techniques, Timeline, Budget, Future Scope, Expected Outcomes, and Conclusion. Use confident and persuasive language, professional terminology, and avoid unsupported claims. Include only statements that have strong support.

Expected Output: A professional proposal written in a persuasive style with clear facts and strong supporting points.

Comparison Note: The Improved version improves the structure, while the Professional version improves the writing style and makes the proposal more convincing and reliable.

Writing a Professional Email

Purpose: To write a clear, professional email that delivers the intended message in the right tone for the target audience.

1. Basic Prompt

Write a professional email about budget extension.

Expected Output: A short and general email without much detail.

2. Improved Prompt

Write a professional email to my manager about budget extension. Give a proper subject line, greet the manager, purpose of the email, key details/points, proper recommendations/steps, and a professional sign-off.

Expected Output: A complete email with a subject, greeting, main message, recommendations, and closing.

3. Professional Prompt

Write a professional email to my manager about budget extension. Give a proper subject line, proper greeting, explain the purpose of the email in the first paragraph. Explain why the current budget is not enough using specific reasons or numbers (already given). Mention the exact budget increase needed, request a reply before a specific date, keep the email under 150 words, and end with a professional sign-off.

Expected Output: A short, professional, and well-organized email with a clear request, supporting reasons, and a response deadline.

Comparison Note: The Professional version is much stronger because it adds clear requirements like a word limit, exact budget request, and response deadline instead of only improving the wording.

Analyzing a Dataset

Purpose: To examine a dataset and identify accurate insights, trends, and patterns based only on the available data.

1. Basic Prompt

Analyze the dataset and give all its findings.

Expected Output: A simple list of findings from the dataset.

2. Improved Prompt

Analyze the given dataset. Tell how many total rows and columns there are, give a summary of what the data contains, identify trends and patterns, check whether any outliers are present, and provide useful insights about the dataset.

Expected Output: A structured analysis with row and column counts, basic trends, and outlier detection.

3. Professional Prompt

Analyze the given dataset as a Data Analyst. First provide an overview of the dataset, including its structure and contents. Perform a complete dataset assessment by identifying missing values, duplicate records, inconsistent or missing information, and data quality issues. Describe the preprocessing steps required before analysis, including handling missing values, duplicates, encoding categorical variables, feature engineering (if required), and outlier detection. After preprocessing, analyze trends and patterns using appropriate statistical methods. Generate useful business insights that support decision-making. Identify whether outliers exist, explain their impact, remove or appropriately handle them, and recommend the best outlier treatment method. Use suitable visualizations (bar charts, histograms, box plots, scatter plots, correlation heatmaps, etc.) to support your analysis. Base every conclusion strictly on the dataset without making unsupported assumptions.

Expected Output: A complete data analysis report including data quality checks, preprocessing steps, statistical analysis, outlier handling, and recommended visualizations.

Comparison Note: This is one of the strongest prompts because it clearly explains every step of the analysis and tells the AI exactly what to do without making assumptions.

Creating Social Media Captions

Purpose: To create short, engaging, and platform-appropriate social media captions that match the brand's style and encourage audience interaction.

1. Basic Prompt

Write a caption on friendship for my Instagram.

Expected Output: A short friendship caption without hashtags.

2. Improved Prompt

Write a caption on friendship for my Instagram, make it funny and lovable, also use hashtags.

Expected Output: A fun and engaging caption with hashtags.

3. Professional Prompt

Write a caption for my post which is a picture of me and my friend. Write something emotional and lovable. Total length must be between 50–100 words. Add hashtags with the names Ayesha and Abeer and end the caption with a heart emoji.

Expected Output: A 50–100-word emotional caption with the required hashtags (#Ayesha and #Abeer) and ending with a heart emoji.

Comparison Note: The Professional prompt is better because it includes clear instructions like word count, required hashtags, and an ending emoji, making the output easy to check.

Generating Chatbot Responses

Purpose: To generate clear, helpful, and consistent chatbot responses that address customer queries while following company policies.

1. Basic Prompt

Write a chatbot response for a patient when they ask about their health conditions.

Expected Output: A simple reply that may not provide enough guidance.

2. Improved Prompt

Write a chatbot response for patients. Use simple English. Give answers to the point. Don't give personal opinions or recommendations. If

patients ask for recommendations, tell them to contact a healthcare professional.

Expected Output: A safe and easy-to-understand response that advises patients to consult a healthcare professional when needed.

3. Professional Prompt

Write a chatbot response system for patients. Use simple English and keep answers short. Never give personal opinions, diagnoses, or treatment advice. If a patient asks for medical advice, tell them to contact a qualified healthcare professional. If they ask for a doctor, provide the doctor's name, specialty, and available consultation hours. If they want to book an appointment, guide them through the booking process and confirm the appointment details. If a patient reports serious or emergency symptoms, immediately advise them to seek emergency medical care. Always keep the tone calm, helpful, and professional.

Expected Output: A complete chatbot system that can answer general questions, help patients find doctors, assist with appointments, and safely handle emergency situations.

Comparison Note: The Professional version is much better because it can handle different situations like doctor lookup, appointment booking, and emergency cases while keeping the responses safe and professional.